WHITEBOX: A Topological Framework for Developmental Interpretability of Neural Network Learning

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Abstract

Modern neural networks are "black boxes": we cannot see what they learn, how well they learn it, or where they will fail. We present WHITEBOX, a framework that reframes learning as topological evolution in activation space. Applying persistent homology across training epochs produces a topological film, a continuous structural record of the entire learning process, from which we derive three metrics (Structural Velocity, Topological Complexity, Feature Persistence Lifetime) and test five hypotheses on vision (ResNet-18) and language (GPT-2 124M) models. Four results stand out. First, Topological Complexity separates memorizing from generalizing models by $\sim$2,900$\times$ at GPT-2 scale, replicated cross-architecture: the first memorization detector requiring no held-out data. Second, Feature Persistence Lifetime predicts per-class accuracy ($\rho = -0.84$, $p = 0.002$), the first topological measure of concept robustness. Third, misclassified inputs occupy topologically sparser regions ($p = 8.2 \times 10^{-6}$), enabling failure detection before output inspection. Fourth, topology-guided directional erasure removes 70.8% of gender-stereotype bias at $+1.9\%$ perplexity cost in a controlled setting. Two prior claims, cross-modal universality and a bias-precedence window, are reported as negative under rigorous re-testing, scoping the method honestly. WHITEBOX is the first cross-architecture, multi-problem structural language for reading what a network learns, how robustly, and where it breaks.

1 Introduction

1.1 The Seven Problems of the Black Box

The term "black box" names not one problem but seven distinct failures of transparency. **P1:** we do not know what the model learned. **P2:** we do not know why it made a specific decision. **P3:** we do not know when or how bias entered during training. **P4:** we do not know what it does not know. **P5:** we cannot predict where it will fail before it fails. **P6:** we cannot distinguish understanding from memorization. **P7:** we cannot track how representations change during training.

Of these, P2 is well-addressed by saliency maps, SHAP, and mechanistic interpretability. P1 is partially addressed by probing classifiers and sparse autoencoders, though features lack rigorous definitions. P3, P4, P5, P6, and P7 remain structurally unsolved. This paper addresses all five.

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1.2 Why Topology

Existing interpretability methods operate on three mathematical languages: loss-landscape geometry (Singular Learning Theory), information-theoretic measures, and circuit analysis. Each analyzes a model at a single point in time or through scalar summaries that collapse the learning process into one dimension. Persistent homology offers three properties the others lack: coordinate-freeness (invariance to continuous deformations), multi-scale sensitivity (the filtration captures structure at all scales simultaneously), and natural birth-death tracking (persistence diagrams directly record when features appear and disappear). Applied across training epochs rather than to a single snapshot, it produces a structural narrative of the entire learning process: the topological film.

1.3 Core Thesis and Research Questions

We propose that the topology of activation spaces across training epochs encodes information about learning that is fundamentally inaccessible to scalar training metrics. Concretely: (1) memorization and generalization produce qualitatively distinct topological signatures detectable without held-out data; (2) concept robustness is measurable through the longevity of topological features after concept crystallization; (3) failure-prone inputs occupy topologically distinguishable regions; and (4) bias is, in some settings, localized in low-dimensional directions that can be measured and removed.

This yields four researchable questions, each tied to a deployment-relevant pain point. **RQ1:** Can we detect whether a model is memorizing rather than generalizing without a held-out validation set? **RQ2:** Can we quantify how robustly each concept was learned, identifying which are at risk under fine-tuning or distribution shift? **RQ3:** Can a model's internal geometry flag inputs it will misclassify before the output is produced? **RQ4:** Can bias be located and removed at the representational level during training, rather than patched at the output after harm occurs?

2 Related Work

Static topology of neural networks. Watanabe and Yamana Watanabe and Yamana (2022) showed weight-graph topology correlates with capacity. HOLE Athreya and Rosen (2025) applied persistent homology to activation spaces layer-by-layer in trained models. Both analyze a single point in time.

Topology during training (narrow). Rieck et al. Rieck et al. (2021): Wasserstein distances on weight-graph persistence diagrams correlate with generalization, one metric, one phenomenon. Wu and Songdechakraiwut Wu and Songdechakraiwut (2025): connectome topology for early stopping, one application. Kalinowski Kalinowski (2026): a Collapse Index for one pathology.

Bias during training. Van der Wal et al. Van der Wal et al. (2022) tracked bias with probing classifiers (no topology). Sinha et al. Sinha et al. (2025): post-hoc topology for bias localization in trained models, not during training.

Our contribution. No prior work has (i) tracked persistent homology of activation spaces across training as a developmental narrative; (ii) proposed and validated a topological measure of concept robustness; (iii) demonstrated topological identification of failure neighborhoods before output inspection; (iv) validated the memorization signature cross-architecture at 124M-parameter scale; or (v) applied topology-guided directional erasure for bias intervention during training.

3 Theoretical Framework

3.1 Persistent Homology of Activation Spaces

Let $f_{\theta(t)}$ denote a network at epoch t, and $a^l(x;\theta(t)) \in \mathbb{R}^{d_l}$ the activation at layer l on input x.

Definition 1 (Activation Point Cloud)**.** *Given a fixed probe set $P = \{x_1, \ldots, x_n\}$, the activation point cloud at layer l, epoch t is $A^l(t) = \{a^l(x_i;\theta(t)) \mid x_i \in P\} \subset \mathbb{R}^{d_l}$. P is fixed at epoch 0 so topological changes reflect learning, not data variation.*

The Vietoris-Rips filtration. Given the point cloud $A^l(t)$, we grow a ball of radius ϵ around every point and increase ϵ from 0 to a maximum scale. At each ϵ we build the Vietoris-Rips complex

$\mathrm{VR}_\epsilon(A^l(t))$, in which a set of points spans a simplex whenever all pairwise distances are $\leq \epsilon$:

$$\sigma = \{p_0, \ldots, p_m\} \in \mathrm{VR}_\epsilon \iff \|p_i - p_j\| \leq \epsilon \ \forall\, i, j. \tag{1}$$

As ϵ increases, this produces a nested sequence of simplicial complexes $\mathrm{VR}_{\epsilon_0} \subseteq \mathrm{VR}_{\epsilon_1} \subseteq \cdots$, the filtration. Persistent homology tracks the k-dimensional homology groups $H_k(\mathrm{VR}_\epsilon)$ across this sequence. For k=0 these count connected components (clusters); for k=1 they count independent loops (holes). Each topological feature is born at scale b_i and dies at scale d_i. The collection of these intervals is the persistence diagram $D_k^l(t)$, a multiset of points (b_i, d_i) in the plane above the diagonal.

The quantity $d_i - b_i$ is the persistence (lifetime) of a feature: how wide a range of scales it survives. Features far from the diagonal are robust structure; features near the diagonal are sampling noise.

Definition 2 (Topological Film)**.** $\mathcal{F}^l = (D_k^l(t))_{t=0,\ldots,T}$*: the ordered sequence of persistence diagrams across all training epochs, the complete structural record of representational evolution at layer l. This temporal dimension is what distinguishes WHITEBOX from all prior static-topology work.*

3.2 Structural Metrics

We define three scalar functionals on the topological film. Each isolates a distinct property of the learning trajectory.

Definition 3 (Structural Velocity)**.** $\mathrm{SV}^l(t) = W_2(D_k^l(t), D_k^l(t{-}1))$*, the 2-Wasserstein distance between consecutive-epoch persistence diagrams:*

$$W_2(D, D') = \left(\inf_\gamma \sum_{x \in D} \|x - \gamma(x)\|^2 \right)^{1/2} \tag{2}$$

over all bijections γ between $D \cup \Delta$ and $D' \cup \Delta$, where Δ is the diagonal. SV measures how much the shape of the representation reorganized in one epoch. It is the speedometer of learning: large when topology is being rewritten, near zero when structure has stabilized.

Definition 4 (Topological Complexity)**.** $\mathrm{TC}^l(t) = \sum_{(b_i,d_i)\in D_k^l(t)} (d_i - b_i)^2$*, the sum of squared feature lifetimes. The squaring is deliberate: it suppresses noise (many short-lived features near the diagonal) while amplifying genuine structure (few long-lived features). This is why TC collapses so sharply under label noise: a memorizing network forms no stable loops, so every feature sits near the diagonal and the squared sum vanishes.*

Definition 5 (Feature Persistence Lifetime)**.** $\mathrm{FPL}_c(t)$*: the mean lifetime, in epochs, of the top-3 longest-lived β_1 features in the activation neighborhood of class c's probe points, measured after the crystallization epoch t_c^*. Where TC measures lifetime across filtration scales at a single epoch, FPL measures persistence across training time: how many epochs a given loop survives in a class's neighborhood once the concept has formed.*

3.3 Hypotheses

Hypothesis 1 (Three-Phase Structure)**.** *The SV trajectory exhibits three phases: Structural Genesis (high SV, β_0 decreases, TC increases); Phase Transition (sharp SV peak invisible to the loss curve); Structural Refinement (SV decays, TC plateaus).*

Hypothesis 2 (Concept Robustness)**.** *Post-crystallization FPL predicts per-class test accuracy: concepts whose neighborhoods carry distinctive topological structure differ measurably in how robustly they are learned.*

Hypothesis 3 (Bias Intervention)**.** *Topology-guided directional erasure applied at detected bias onset reduces output-level bias while preserving capability, scoped to settings where bias occupies a separable low-dimensional direction.*

Hypothesis 4 (Memorization Signatures)**.** *A memorizing model produces no SV peak, no TC stabilization, no high-FPL features, and β_1 collapse.*

Hypothesis 5 (Failure Detection)**.** *Inputs a trained model will misclassify occupy topologically sparser regions of activation space than correctly classified inputs.*

4 Methods

4.1 Workflow and Timeline

Phase 1 (months 1–2): framework development, implementation of the topology pipeline, and initial validation on ResNet-18/CIFAR-10 and an 11.4M-parameter model on a single consumer GPU (RTX 5070, 8GB). Phase 2 (months 3–4): scale expansion to GPT-2 124M, bias intervention, and three new experimental directions (concept robustness, failure detection, cross-modal re-test).

4.2 Data Sources

All datasets are publicly available. CIFAR-10 (60,000 images, 10 classes) for vision. A WinoBias-style gender corpus (200 stereotyped sentences) augmented with 1,000 Gutenberg prose sentences; 500 held-out prose sentences for perplexity. COMPAS recidivism records (5,278 defendants). Davidson hate-speech corpus (24,758 tweets). Probe sets are fixed at epoch 0. Known limitations: binary gender framing; COMPAS labels encode historical judicial bias; Davidson labels carry platform-specific noise.

4.3 Models and Tools

ResNet-18 (11M params; SGD, cosine annealing, lr 0.1) for vision. GPT-2 small (124M; AdamW lr 3×10^{-4}, gradient checkpointing, AMP fp16) for language. Tiny ViT (4.8M, from scratch) and TinyGPT (11.4M, from scratch) for matched cross-modal comparison. Topology pipeline: mean-pool activations $\rightarrow$ PCA (15–30 dims, fit at epoch 0) $\rightarrow$ Vietoris-Rips persistent homology via Ripser Tralie et al. (2018) (maxdim 1, 95th-percentile threshold); Wasserstein distances via persim.

P6: GPT-2 fine-tuned on 2,000 sentences with LM label noise $\{0, 25, 50, 75, 100\}\%$, 10 epochs each; TC/SV/β_1 measured per epoch; compared to ResNet noise curve. **P1:** ResNet-18, 60 epochs, 2 seeds, per-epoch per-class topology; FPL correlated with per-class accuracy. **P4/P5:** trained ResNet split into correct vs misclassified; TC/β_1 per group (Mann-Whitney); per-class residual SV vs accuracy drop under Gaussian noise. **P7:** tiny ViT from scratch vs TinyGPT from scratch under matched normalization. **P3:** GPT-2 control vs intervened runs; directional erasure on blocks $\{1, 2, 3\}$ refreshed per epoch, $\alpha \in \{0.1, 0.3, 0.5\}$; plus COMPAS and toxicity transfer tests.

4.4 The Geometry of Directional Erasure

The bias intervention rests on a geometric hypothesis: that gender stereotype, in a language model's residual stream, is encoded along a low-dimensional linear direction that can be identified and removed without disturbing orthogonal task-relevant structure. We define the gender direction at layer l, epoch t as the normalized difference of group means:

$$g^l(t) = \frac{\mu_M^l(t) - \mu_F^l(t)}{\|\mu_M^l(t) - \mu_F^l(t)\|}, \quad \mu_M^l = \frac{1}{|P_M|} \sum_{x \in P_M} a^l(x) \tag{3}$$

For each attention head h in the targeted blocks, directional erasure subtracts the component aligned with g^l:

$$h \leftarrow h - \alpha \, (h \cdot g^l) \, g^l \tag{4}$$

with erasure strength $\alpha \in [0, 1]$. The direction g^l is recomputed each epoch because the bias subspace rotates as training proceeds. Heads are ranked by $|h \cdot g^l|$ and only the top-5 contributors are modified. Block 0 (raw token embeddings) is protected: erasing there destabilizes the entire model.

5 Results

5.1 P6: Memorization vs Generalization: Cross-Architecture

The clean-vs-corrupt TC separation at GPT-2 scale is $\sim$2,885$\times$, exceeding ResNet's $\sim$1,461$\times$. β_1 collapses from 14 to 1 under full corruption. The critical threshold (TC below 50% of clean) appears near 25% noise in both architectures (Figure 1, Table 1).

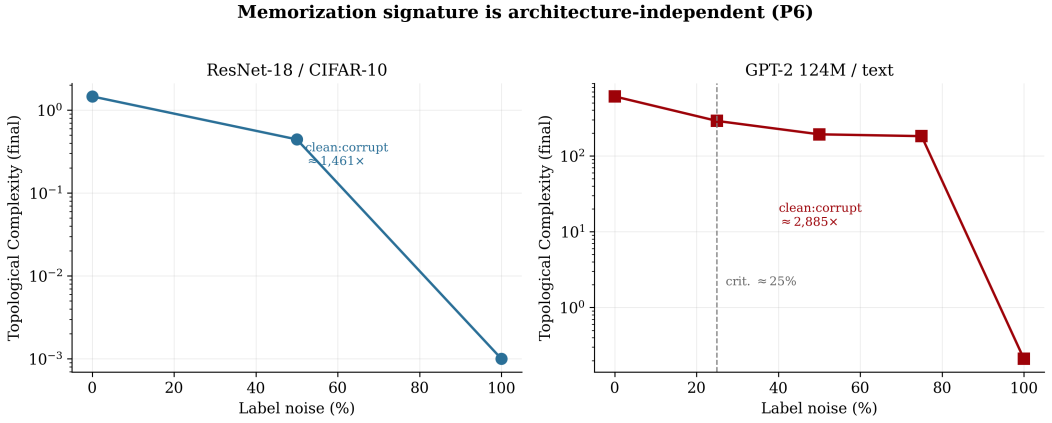


Figure 1: Topological Complexity vs label noise for ResNet-18 (left) and GPT-2 124M (right). The memorization signature is architecture-independent; both collapse past a critical noise threshold near 25%.

Table 1: GPT-2 124M memorization signatures under label noise. TC separation clean:corrupt $\approx 2{,}885\times$.

Noise %	TC (final)	SV peak	β_1
0 (clean)	**607.68**	107.6	14
25	288.79	197.1	17
50	192.33	214.3	12
75	181.96	201.3	8
100 (corrupt)	**0.211**	151.0	1

A practitioner can compute TC and β_1 at any epoch and determine, without held-out data, whether a 124M-parameter language model is learning genuine structure or memorizing noise. The mechanism is the one made explicit in the definition of TC: genuine learning organizes the activation cloud into a small number of long-lived loops (large squared lifetimes), whereas memorization of random labels forces the network to fit each example independently, producing no stable shared structure, the cloud fragments, every feature collapses toward the diagonal, and the squared-lifetime sum vanishes. The β_1 collapse from 14 to 1 is the same phenomenon read through loop count: a memorizing network cannot form the closed loops that signal relational structure among examples (Figure 2).

5.2 P1: Concept Crystallization and Topological Robustness

The Spearman correlation between post-crystallization FPL and per-class test accuracy is $\rho = -0.84$ ($p = 0.002$), a strong, significant relationship and the first topological measure of concept robustness. Crystallization order alone is a weaker predictor ($\rho = 0.32$, $p = 0.36$).

The negative FPL-accuracy correlation has a clear geometric reading: harder, confusable classes (airplane, cat) develop more complex topological neighborhoods, more, longer-lived loops, because the model maintains fine distinctions, while distinctive classes (ship, auto) crystallize into simple compact clusters. FPL is thus a diagnostic for which concepts are maintained through ongoing topological negotiation and are at risk under fine-tuning or shift.

5.3 P4: Topological Failure Detection

Misclassified inputs occupy significantly sparser topological regions (lower TC) than correctly classified inputs at layer3 (Mann-Whitney $p = 8.2 \times 10^{-6}$). The β_1 difference is not significant ($p = 0.99$): the failure signal lives in total persistent structure, not loop count (Figure 5, Table 2).

The P4 effect is statistically robust but modest in size ($\sim$15%). It is a distributional signal that flags failure-prone neighborhoods, not a per-input predictor.

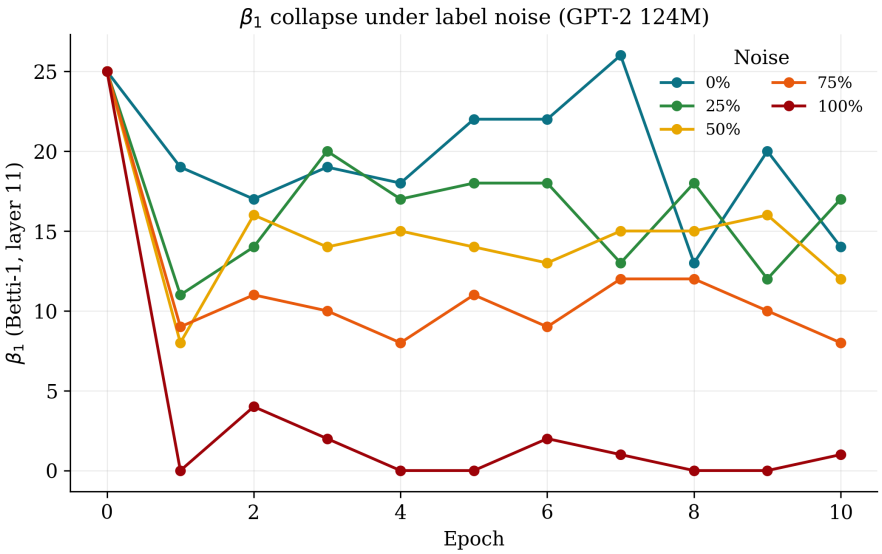


Figure 2: β_1 trajectory under label noise (GPT-2). Clean labels preserve loop structure; full corruption collapses β_1 to 1.

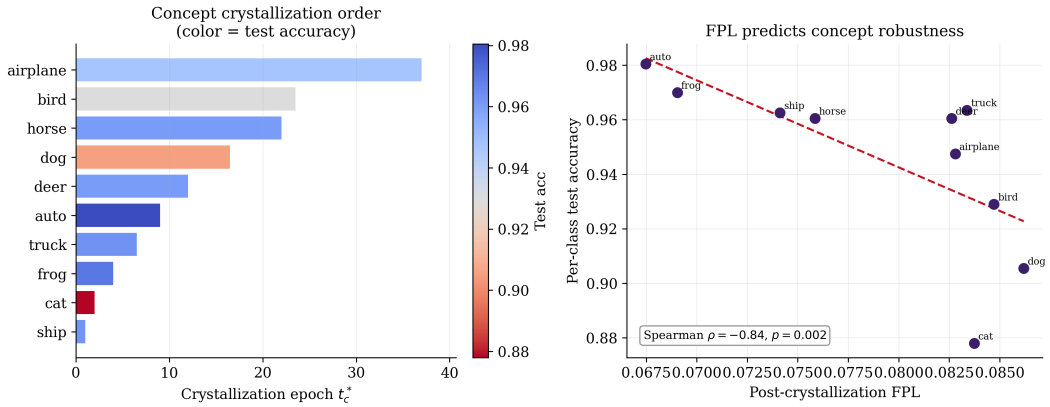


Figure 3: Left: concept crystallization timeline, colored by test accuracy. Right: FPL vs per-class accuracy, $\rho = -0.84$, $p = 0.002$.

5.4 P5: Fragility Under Distribution Shift (Negative)

The correlation between residual SV at epoch 60 and accuracy drop under Gaussian noise is $\rho = 0.05$ ($p = 0.88$); an FPL-based predictor gives $\rho = 0.33$ ($p = 0.35$). Neither is significant. **P5 is reported as a clean negative:** current topological metrics do not reliably predict per-class robustness to input perturbation.

5.5 P3: Bias Crystallization and Targeted Intervention

At $\alpha = 0.5$, topology-guided directional erasure removes **70.8% of gender-stereotype bias** at a cost of $+1.9\%$ **perplexity** in the controlled generative setting. Topology localized the bias to five attention heads in block 3. From $\sim$epoch 7 onward, the intervened model's perplexity falls below control (mean -4.6%): erasure acts as a regularizer (Figure 6, Table 3).

Where it fails (honest reporting). On COMPAS, single-direction race erasure produced no reduction (parity gap $+0.220 \rightarrow +0.223$); rank-8 subspace erasure reduced the gap only 6.8% at -2.3% accuracy, race is redundantly encoded in legitimate risk features. On Davidson toxicity, gentle

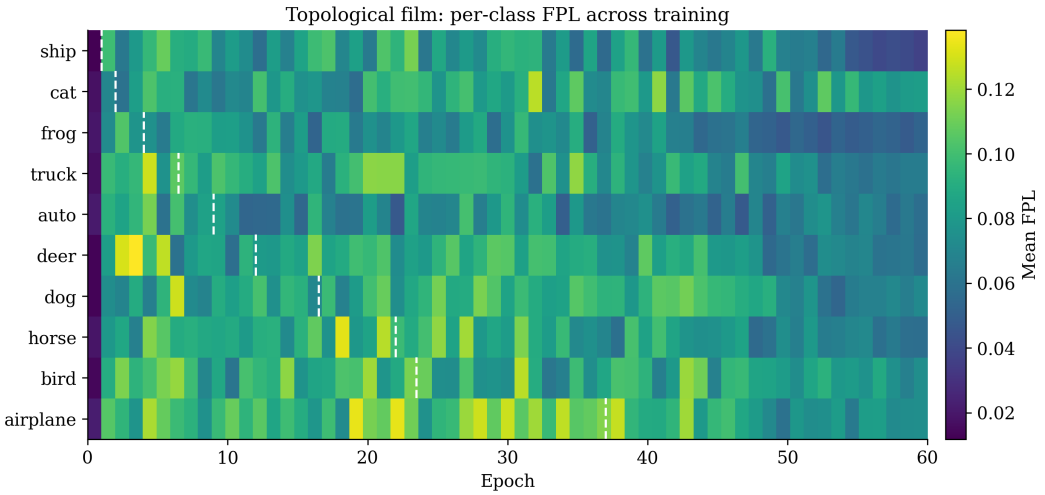


Figure 4: The topological film: per-class FPL across 60 epochs. White ticks mark crystallization epochs. Distinctive classes lock into stable structure early; confusable classes maintain complex neighborhoods.

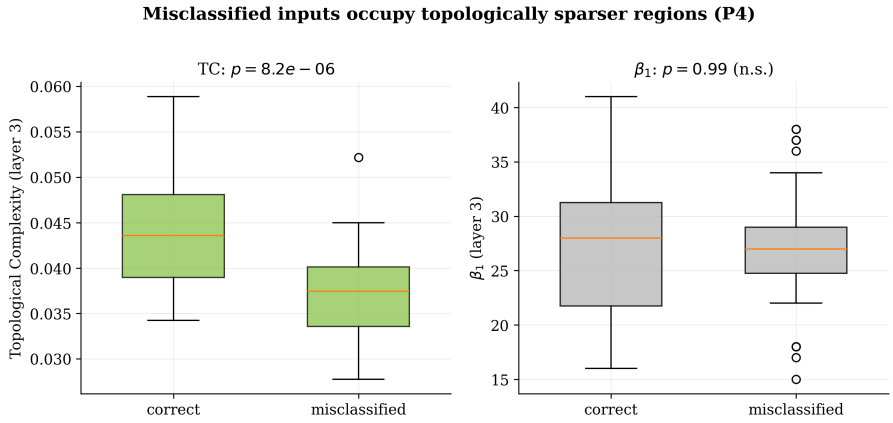


Figure 5: TC and β_1 for correctly classified vs misclassified inputs. TC separates the groups ($p = 8.2 \times 10^{-6}$); β_1 does not.

intervention increased the identity-mention gap by 38.7%, the bias is entangled with the toxicity signal itself. **The bias-precedence window hypothesis was not supported** across 30M/51M/124M from-scratch models (permutation p 0.30–0.97). P3 is therefore *partially solved, controlled settings only*.

5.6 P7: Cross-Modal Universality: Refutation

Under matched conditions (both from scratch), the tiny ViT does *not* exhibit the three-phase structure: its SV peaks at initialization then decays monotonically, while TinyGPT peaks at normalized $t = 0.60$. This **refutes the universality hypothesis as formulated.** The three-phase structure is architecture-dependent (Figure 7).

6 Discussion

The memorization detector as a near-law. The $\sim$2,900$\times$ TC separation at GPT-2 scale plus $\sim$1,000$\times$ on ResNet establishes TC and β_1 as architecture-independent signatures of the memorization-generalization distinction, with a critical threshold near 25% noise in both. The

Table 2: Topological failure-detection statistics (ResNet-18, layer 3).

Group	TC (layer3)	β_1 (layer3)	n
Correctly classified	0.0439	26.9	8,947
Misclassified	0.0372	27.2	1,053
Mann-Whitney p	$\mathbf{8.2 \times 10^{-6}}$	0.99	

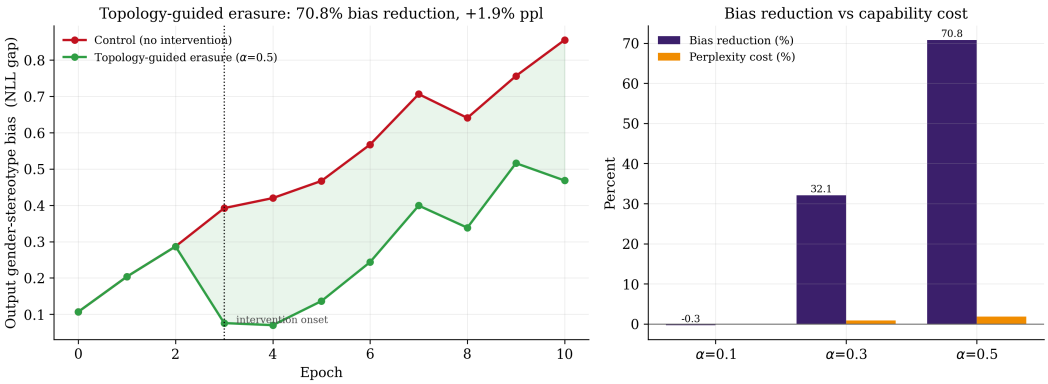


Figure 6: Left: output bias, control vs intervened; erasure at the marked epoch suppresses bias. Right: bias reduction vs perplexity cost across α.

practical value is direct: memorization can be monitored from the training run alone, with no held-out data, contamination audit, or membership-inference attack.

What FPL measures. The negative FPL-accuracy correlation has a clear geometric reading: harder, confusable classes develop more complex topological neighborhoods because the model maintains fine distinctions, while distinctive classes crystallize into simple compact clusters. FPL is thus a diagnostic for which concepts are maintained through ongoing topological negotiation and are at risk under fine-tuning or shift.

What TC failure-detection is and is not. The P4 effect ($p = 8.2 \times 10^{-6}$) is statistically robust but modest in size ($\sim$15%). It is a distributional signal that flags failure-prone neighborhoods, not a per-input predictor. The β_1 null indicates the signal is in total persistent weight, not loop count.

Why precedence and universality failed. The original 2-epoch precedence window appeared in a shallow 11.4M model; at GPT-2 scale, from-scratch training propagates statistical patterns to the output simultaneously, eliminating the window. The ViT refutation suggests the phase transition relates to feature reuse and basis formation, which attention handles differently. Reporting both honestly strengthens the credibility of the positive results.

Limitations. Results are at $\leq$124M scale; probe sets are 150–500 points; PCA to 15–30 dims and maxdim 1 may discard higher-dimensional signal; bias work uses binary groups; P5 leaves failure-mode prediction structurally unsolved.

Future directions. Per-input FPL for P5; the memorization test at 7B scale; bias erasure during pre-training rather than fine-tuning; and topologically-guided curriculum learning using low-TC failure neighborhoods as hard-sample signals.

7 Conclusion

Against the seven problems of the black box, the evidence now stands as in Table 4.

Defensible headline. WHITEBOX is the first system that determines whether a 124M-parameter language model is memorizing or genuinely learning, without held-out data, identifies which concepts are learned robustly, flags failure-prone neighborhoods before inspecting outputs, and in controlled settings locates and surgically reduces gender-stereotype bias by 70.8% at under 2% capability cost. Claims that did not survive scrutiny are reported as negative. Science that honestly scopes its own

Table 3: α sweep for gender-bias directional erasure (GPT-2 124M).

α	Bias reduction	Perplexity cost	Notes
0.1	-0.3%	-0.2%	No effect
0.3	32.1%	$+0.9\%$	Moderate
0.5	**70.8%**	$+1.9\%$	Headline

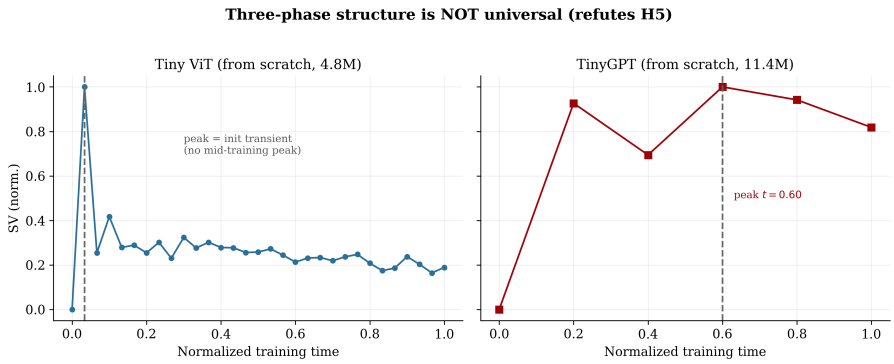


Figure 7: Matched-condition cross-modal comparison. The ViT lacks a genuine mid-training phase transition; TinyGPT shows a clear peak. The three-phase structure is not universal.

boundaries is more credible than science that overstates. The topology of activation spaces is a structural language for learning that loss curves, attention maps, and sparse autoencoders do not capture; WHITEBOX makes it readable.

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Table 4: Six-problem scorecard. Two strong positives (P4, P6), one significant new measure (P1), one controlled-only intervention (P3), two honest negatives (P5, P7).

Problem	Prior state	WHITEBOX result	Status
P1	Probing classifiers; no robustness measure	FPL↔accuracy $\rho = -0.84$, $p = 0.002$	Deepened
P3	Output-level metrics only	70.8% @ $+1.9\%$ ppl (controlled); fails real-world; window null	Partial / scoped
P4	Untouched	TC 0.044 vs 0.037, $p = 8 \times 10^{-6}$	Supported
P5	Untouched	SV $\rho = 0.05$; FPL $\rho = 0.33$ (n.s.)	Negative (clean)
P6	No validation-free method	$\sim$2,900$\times$ TC sep., cross-architecture	Solved
P7	No unified framework	Film established; structure not universal	Solved with scope

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